

An observable can miss the slow mixing mode

Disproof of Conjecture 00000003429

ziangni-sys

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Scope. Both source language versions identify the autocorrelation decay of observables with the mixing spectral radius and propose lag regression to estimate that radius. An individual observable need not detect the slowest nonconstant mode. We give a strictly positive, lazy, reversible finite-state chain and a nonzero centered observable whose exact exponential decay factor is strictly smaller than the mixing spectral radius. The example refutes the blanket equality for observables. It does not refute a worst-case upper bound or an equality obtained by taking the supremum over all observables, nor a claim restricted to observables having nonzero projection onto the slowest mode.

A genuine Markov chain. On the state space $\{0, 1, 2\}$, use the transition matrix

$$P = \begin{pmatrix} 5/8 & 1/8 & 1/4 \\ 1/8 & 5/8 & 1/4 \\ 1/4 & 1/4 & 1/2 \end{pmatrix}, \quad \pi = (1/3, 1/3, 1/3).$$

Every entry is strictly positive and each row sums to one. Thus the chain is irreducible and aperiodic. Its diagonal entries are at least $1/2$, so it is lazy. Symmetry gives both stationarity $\pi P = \pi$ and detailed balance $\pi_i P_{ij} = \pi_j P_{ji}$.

The actual centered restriction. The centered space is $H_0 = \{v : \sum_i \pi_i v_i = 0\}$. Set $g = (1, -1, 0)$ and $f = (1, 1, -2)$. These vectors form a basis of H_0 , and direct multiplication gives

$$Pg = \frac{1}{2}g, \quad Pf = \frac{1}{4}f.$$

For completeness, the coordinate map and its inverse are explicit:

$$E(a, b) = (a + b, -a + b, -2b), \quad a = \frac{v_0 - v_1}{2}, \quad b = \frac{v_0 + v_1}{2} \quad (v \in H_0).$$

Thus E is a linear bijection onto H_0 and

$$PE = ED, \quad D = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/4 \end{pmatrix}.$$

The spectrum of this restriction is exactly $\{1/2, 1/4\}$, as also follows from $\det(zI - D) = (z - 1/2)(z - 1/4)$. The same diagonalization holds on complexifying H_0 , so no nonreal spectral values arise. The mixing spectral radius, the largest modulus on the centered restriction, is therefore

$$\rho_{\text{mix}} = \frac{1}{2}.$$

Theorem 1. *For the nonzero centered observable $f = (1, 1, -2)$, the normalized stationary autocorrelation is $C_f(n) = 4^{-n}$. Its exponential decay factor and every consecutive-lag regression ratio equal $1/4$, whereas $\rho_{\text{mix}} = 1/2$.*

Proof. Induction gives $P^n f = 4^{-n} f$. In stationarity, the Markov property gives

$$\text{Cov}_\pi(f(X_0), f(X_n)) = \sum_i \pi_i f_i (P^n f)_i = 4^{-n} \frac{1 + 1 + 4}{3} = 2 \cdot 4^{-n},$$

because $\sum_i \pi_i f_i = 0$. In particular $\text{Var}_\pi(f) = 2 > 0$, so normalization gives $C_f(n) = 4^{-n}$. For all $n \geq 0$,

$$\frac{C_f(n+1)}{C_f(n)} = \frac{1}{4}, \quad |C_f(n+1)|^{1/(n+1)} = \frac{1}{4}.$$

Hence the positive-root exponential-rate limit is exactly $1/4$, strictly smaller than $1/2$. In logarithmic conventions the rates are $\log 4$ and $\log 2$, likewise unequal. $\square$

Why the spectral information is lost. Under the stationary inner product, $\langle f, g \rangle_\pi = 0$. This observable has no component in the eigenvalue- $1/2$ direction. There is no defect in positivity, laziness, reversibility, centering, or variance: it simply probes a different nonconstant eigenspace. Lag regression cannot recover an unobserved mode from this autocorrelation.

Lean coverage. The project defines the actual transition matrix and stationary weights, proves positivity, stochasticity, laziness, stationarity and detailed balance, and uses explicit coordinates to identify the entire centered restriction. It computes its spectrum and spectral radius via matrix invertibility. Autocovariance is the actual stationary finite-sum formula using powers of P . Its exact value, normalized correlation, all lag ratios and the root-rate limit are proved, ending in the inequality between the two rates. No eigenvalue, covariance identity, or decay-rate certificate is assumed as a hypothesis.